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matter**

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COS 330 – Practical 3

Malware Classification with Static Analysis

Due date: 24th September 2025 @ 11:00 am

Background

Malware continues to pose one of the most significant threats to information systems worldwide. Unlike benign software, which is developed with the intent to serve legitimate purposes, malware is designed to disrupt, damage, or gain unauthorized access to systems and data. Attackers frequently employ various obfuscation techniques to hide malicious behaviour, making it increasingly difficult to detect using traditional signature-based methods.

Static analysis is a core technique in malware research and detection. Unlike dynamic analysis, which executes a program to observe its behaviour, static analysis inspects the binary or code without execution. This allows security researchers to uncover suspicious strings, metadata, headers, imports, and other structural features that reveal insights about potential malicious intent. Tools such as PESTudio provide automated extraction of file characteristics, while frameworks like YARA enable analysts to craft detection rules that classify files based on unique patterns.

In this assignment, you will perform malware classification by combining static analysis with rule-based detection. You will work with both benign and malicious executables from trusted datasets, extract relevant features, and design YARA rules that can identify malware families or characteristics.

Instructions

For this practical, you have to implement the tasks below as well as **write a report on your findings**. Be sure to document all you have done, including screenshots as proof. You are restricted to using **YARA Rules**, but you **may use any dataset to obtain the samples**.

Scenario

You have been contracted by CyberDef Labs, a security research organization, to develop a lightweight malware classification system that relies on static analysis. **The goal is to detect malicious binaries** before they are executed, reducing the risk of system compromise. The organization requires you to:

- Build a dataset of benign and malicious samples (at least 5 of each type).
- Extract features from these samples using static analysis methods.

Write YARA rules to classify malware into categories based on the extracted characteristics. Document your methodology, detection performance, and reasoning in a report.

The system is not expected to be flawless but should demonstrate practical malware detection logic and illustrate how static analysis can be applied in real-world security operations.

Task 1 – [15 marks]

Download at least **5 benign executables** and **5 malicious executables** from trusted sources (e.g., **VirusShare**, **theZoo**, **MalwareBazaar**, **PortableApps**, etc).

Clearly document where you obtained the samples and how you ensured their integrity (e.g., via **hash verification**).

Organize your dataset into appropriate folders (benign vs malicious).

NOTE: These will be real malware, be sure to use a VM to protect your system.

Task 2 – [35 marks]

Use **PEStudio** or equivalent tools to extract static features from both benign and malicious executables.

Analyse features such as:

- Import tables (DLLs)
- PE headers and sections
- Embedded strings (e.g., suspicious URLs, commands)
- File metadata (timestamps, compiler info)

Present your findings in tabular or graphical form in your report.

Task 3 – [30 marks]

Write at least 3 YARA rules that detect and classify malware types based on your extracted features.

Each rule must:

- Have a clear name and description.
- Specify the conditions under which a file will be classified as malicious.
- Be tested against your dataset (benign and malicious).
- Document the logic and reasoning behind each rule (e.g., why a certain string indicates malicious intent).

Hint: Rules should be specific enough to catch malware without generating excessive false positives.

Bonus – [2 marks]

Any additional or out of the ordinary security features / ML detection performed.

Submission

In your report, include screenshots, the hashes of the samples as well as name, where it was sourced. Include your YARA Rules, and the features you are extracting, and how you extracted them.

Submit your report as a PDF file **ONLY** called uXXXXXX_NAME_P3.pdf to ClickUP before the deadline.

Please upload well in advance, as there will be

ABSOLUTELY NO LATE SUBMISSIONS!!!

TOTAL: 80

